

MD-I Question Bank

Unit No.01- Mechanical Engineering Design Process

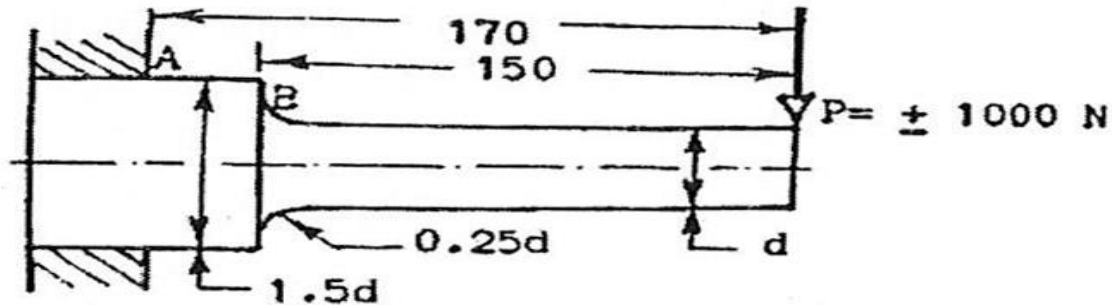
1. Define machine and machine design. Give some examples of machine.
2. Explain various types of loads with examples.
3. What is the factor of safety? Explain their significance in designing various machine elements.
4. Explain factor on which magnitude of factor of safety depends?
5. Explain General /Basic design procedure for a machine design.
6. Explain the design considerations for manufacturing.
7. Explain the design considerations for Forging.
8. Explain the design considerations for assembly
9. Explain the design considerations for machined parts or machining.
10. Explain the design considerations for casting.
11. Discuss any two of the following theories of failure and its applications in design of machine component.
 - i) Max. principal stress theory
 - ii) Max. shear stress theory
 - iii) Distortion energy theory

Unit No. 02-Design of machine elements against Static Load

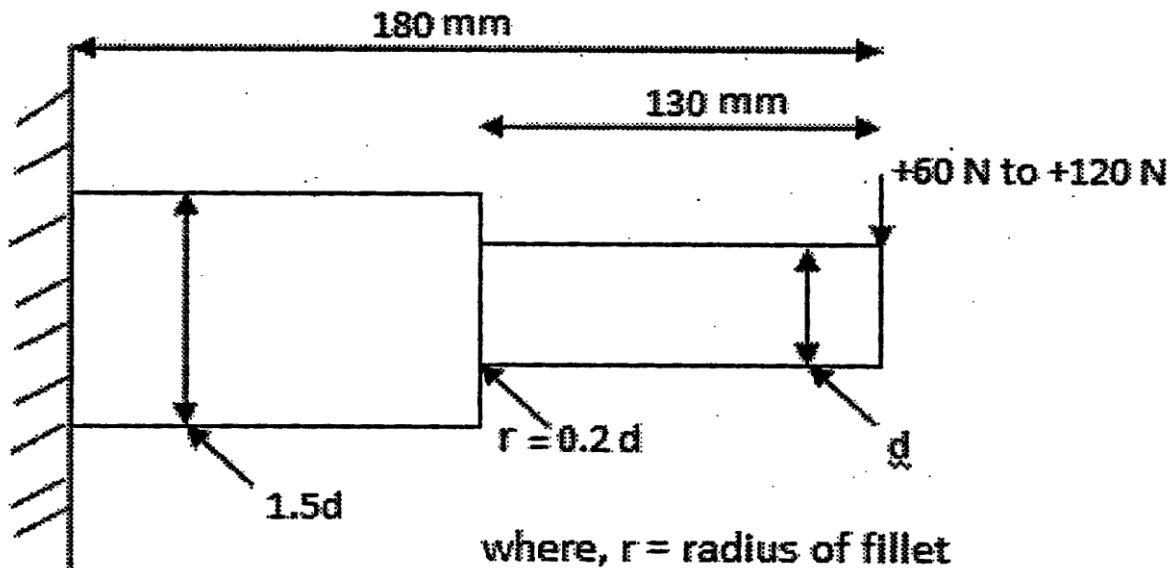
1. Write the Design Procedure of Socket and Spigot Cotter Joint.
2. Write the Design Procedure of Knuckle Joint.
3. Design and draw a cotter joint to support a load varying from 30 kN in compression to 30 kN in tension. The material used is carbon steel for which the following allowable stresses may be used. The load is applied statically. Tensile stress = compressive stress = 50 MPa ; shear stress = 35 MPa and crushing stress = 90 MPa.
4. It is required to design a knuckle joint to connect two circular rods subjected to an axial tensile force of 50 kN. The rods are co-axial and a small amount of angular movement between their axes is permissible. Design the joint and specify the dimensions of its components. Select suitable materials for the parts.

Unit No. 03 - Design against fluctuating loads

1. What are the various causes of stress concentration? Discuss different methods for reducing the effect of stress concentration.
2. Explain notch sensitivity, Endurance limit, Cyclic Stresses.
3. Explain Soderberg and Goodman diagram
4. A cantilever beam made of cold drawn steel 20 C8 ($S_{ut} = 540 \text{ N/mm}^2$) is subjected to a completely reversed load of 1000 N as shown in Fig. The notch sensitivity factor q at the fillet can be taken as 0.85 and expected reliability is 90 percent, determine the diameter (d) of the beam for a life of 10000 cycles. Take $K_a = 0.78$, $K_b = 0.85$, $K_c = 0.897$, $K_t = 1.35$.

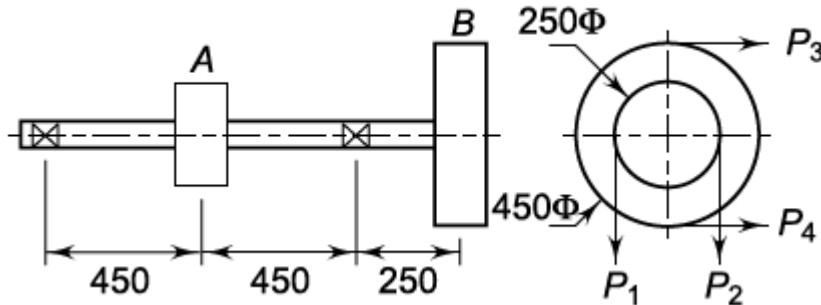


5. A cantilever beam made of cold drawn steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$, $S_{ut} = 630 \text{ N/mm}^2$) is shown in fig. The force acting at the free end varies from + 60 N to + 120 N. The expected reliability is 90% and the factor of safety is 2. The notch sensitivity factor at the fillet can be taken as 0.9. Determine the diameter ' d ' of the beam at the fillet cross-section. Take $K_a = 0.77$, $K_b = 0.85$, $K_c = 0.897$ and $K_t = 1.44$.



Unit No. 04 Design of Shaft Keys and Couplings

1. Discuss the merits and demerits of solid shaft and hollow shaft.
2. A line shaft supporting two pulleys A and B is shown in Fig. 9.5(a). Power is supplied to the shaft by means of a vertical belt on the pulley A, which is then transmitted to the pulley B carrying a horizontal belt. The ratio of belt tension on tight and loose sides is 3:1. The limiting value of tension in the belts is 2.7 kN. The shaft is made of plain carbon steel 40C8 ($S_{ut} = 650 \text{ N/mm}^2$ and $S_{yt} = 380 \text{ N/mm}^2$). The pulleys are keyed to the shaft. Determine the diameter of the shaft according to the ASME code if, $k_b = 1.5$ and $k_t = 1.0$



3. A solid shaft of diameter d is used in power transmission. Due to modification of the existing transmission system, it is required to replace the solid shaft by a hollow shaft of the same material and equally strong in torsion. Further, the weight of the hollow shaft per meter length should be half of the solid shaft. Determine the outer diameter of the hollow shaft in terms of d .
4. It is required to design a square key for fixing a pulley on the shaft, which is 50 mm in diameter. The pulley transmits 10 kW power at 200 rpm to the shaft. The key is made of steel 45C8 ($S_{yt} = S_{yc} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Determine the dimensions of the key. Assume ($S_{sy} = 0.577S_{yt}$)
5. A flexible coupling is used to transmit 15 kW power at 100 rpm. There are six pins and their pitch circle diameter is 200 mm. The effective length of the bush (l_b), the gap between two flanges and the length of the pin in contact with the right hand flange are 35, 5 and 23 mm respectively. The permissible shear and bending stresses for the pin are 35 and 152 N/mm^2 respectively. Calculate: (i) pin diameter by shear consideration; and (ii) pin diameter by bending consideration.
6. It is required to design a split muff coupling to transmit 50 kW power at 120 rpm. The shafts, key and clamping bolts are made of plain carbon steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$). The yield strength in compression is 150% of the tensile yield strength. The factor of safety for shafts, key and bolts is 5. The number of clamping bolts is 8. The coefficient of friction between sleeve halves and the shaft is 0.3.
 - (i) Calculate the diameter of the input and output shafts.
 - (ii) Specify the length and outer diameter of the sleeve halves.
 - (iii) Find out the diameter of clamping bolts assuming that the power is transmitted by friction.
 - (iv) Specify bolt diameter using standard empirical relations.
 - (v) Specify the size of key and check the dimensions for shear and compression criteria.
7. A rigid coupling is used to connect a 45 kW, 1440 rpm electric motor to a centrifugal pump. The starting torque of the motor is 225% of the rated torque. There are 8 bolts and their pitch circle diameter is 150 mm. The bolts are made of steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 2.5. Determine the diameter of the bolts. Assume ($S_{sy} = 0.577S_{yt}$) Assume that the bolts are finger tight in reamed and ground holes.
8. Discuss ASME codes for shaft design.

Unit No. 05 Design of Threaded Joints and Mechanical Springs

1. What are the different types of springs and their applications? Explain in brief helical compression springs.
2. Define the following terms :
 - i) Solid length
 - ii) Free length
 - iii) Spring index
 - iv) Spring stiffness.
3. Write the stress and deflection equations for helical spring.
4. A helical compression spring, made of circular wire, is subjected to an axial force, which varies from 2.5 kN to 3.5 kN. Over this range of force, the deflection of the spring should be approximately 5 mm. The spring index can be taken as 5. The spring has square and ground ends. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1050 N/mm² and modulus of rigidity of 81370 N/mm². The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength. Design the spring and calculate
 - (i) wire diameter;
 - (ii) mean coil diameter;
 - (iii) number of active coils;
 - (iv) total number of coils;
 - (v) solid length of the spring;
 - (vi) free length of the spring;
 - (vii) required spring rate; and
 - (viii) actual spring rate
5. A railway wagon moving at a velocity of 1.5 m/s is brought to rest by a bumper consisting of two helical springs arranged in parallel. The mass of the wagon is 1500 kg. The springs are compressed by 150 mm in bringing the wagon to rest. The spring index can be taken as 6. The springs are made of oil-hardened and tempered steel wire with ultimate tensile strength of 1250 N/mm² and modulus of rigidity of 81 370 N/mm². The permissible shear stress for the spring wire can be taken as 50% of the ultimate tensile strength. Design the spring and calculate:
 - (i) wire diameter;
 - (ii) mean coil diameter;
 - (iii) number of active coils;
 - (iv) total number of coils;
 - (v) solid length;
 - (vi) free length;
 - (vii) pitch of the coil;
 - (viii) required spring rate; and
 - (ix) Actual spring rate.
6. What are the advantages and disadvantages of welded joints?
7. Explain advantages of riveted joint over welded joints.